

CHAPTER 1

INTRODUCTION

In the ever-evolving landscape of transportation, the imperative to enhance road safety and rider convenience has become increasingly pronounced. Two-wheeled vehicles, ubiquitous in urban environments, demand innovative solutions to address the challenges posed by dynamic traffic scenarios. One such pioneering solution is the development of an Adaptive Cruise Control (ACC) system tailored explicitly for two-wheelers.

The core objective of this project is to design and manufacture an ACC system that goes beyond the conventional applications in four-wheeled vehicles. Adaptive Cruise Control, a sophisticated control mechanism, integrates state-of-the-art sensors, actuators, and intelligent algorithms to facilitate automatic speed adjustment based on the dynamic traffic environment. By bringing this technology to the realm of two-wheelers, we aim to significantly improve rider safety and convenience.

The methodology employed in this endeavor is multifaceted. It commences with a comprehensive study of existing ACC systems, both in the context of four-wheeled vehicles and the unique challenges presented by the dynamics of two-wheelers in diverse traffic conditions. This initial phase sets the foundation for the subsequent steps, which involve the selection and integration of appropriate sensors, the design of a compact and efficient actuator system, and the formulation of a robust control algorithm.

Moving into the design and development phase, our focus lies on achieving seamless integration of sensors, implementing an effective actuator system, and creating a control unit that ensures reliability and compatibility in real-world scenarios. The intricacies of hardware and software integration will be carefully addressed to guarantee optimal performance.

The testing and validation procedures will be rigorous, encompassing simulations, bench testing, and on-road trials. Performance metrics will be meticulously established to evaluate the responsiveness, accuracy, and overall effectiveness of the

designed ACC system. Results and subsequent discussions will provide a comparative analysis, showcasing the novel contributions of this ACC system to the domain of two-wheeled vehicles.

In conclusion, the successful realization of this project not only presents a functional ACC system for two-wheelers but also signifies a leap forward in integrating cutting-edge technology to improve the safety and comfort of riders in urban traffic scenarios. The ensuing sections of this report will delve into the intricacies of the design and manufacturing processes, detailing the challenges, solutions, and future prospects of this innovative endeavor.

1.1 Problem Statement

1. The Adaptive Cruise Control System is available in four-wheeler and in gear bike.
2. More than 15km distance the Non-gear bike is not comfortable.
3. There is no extremely innovation about ACC in mopeds (Non Gear two wheelers), So in this project we are developing the Adaptive cruise control system for moped.

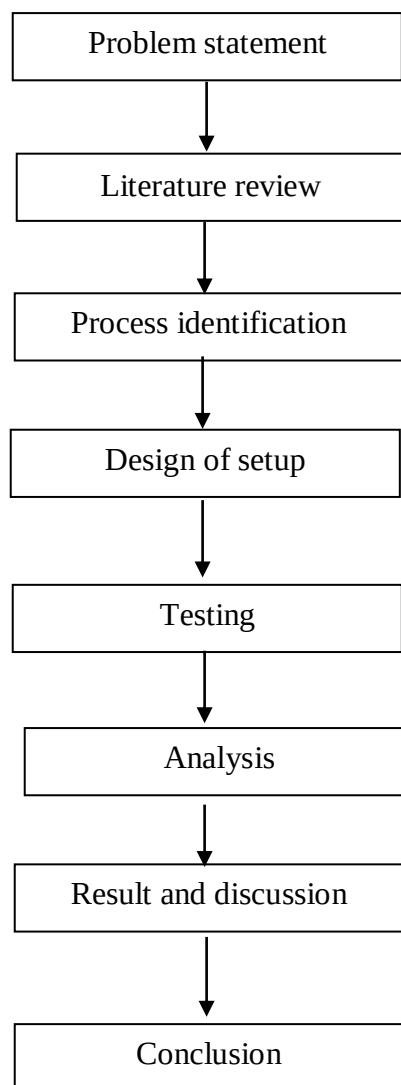
1.2 Objective

1. **Enhanced Riding Comfort:** ACC takes over the task of maintaining a constant speed and following distance, allowing the rider to focus more on the surrounding environment, thus reducing fatigue during long rides.
2. **Fuel Efficiency:** ACC optimizes speed control by avoiding unnecessary acceleration and deceleration, leading to improved fuel efficiency and reduced emissions.
3. **Technology Showcase:** Implementing ACC in two-wheelers serves as a showcase of technological innovation, demonstrating the industry's commitment to improving rider safety and comfort.

1.3 Scope

1. Customization for Two-Wheelers
2. Add new feature in a moped, by means of electronic device.
3. Market Demand and Consumer Acceptance.

1.4 Methodology



CHAPTER 2

LITERATURE REVIEW

In this project the main aim is to develop a device which is use to improve rider comfort. So we develop a device known as ACC (Adaptive cruise control), which is use to maintain vehicle's speed as per our desire speed, by using a locking mechanism. The proposed work that has been done mainly on the basis of the literature review is explained below.

2.1) ACC to avoid and reduce accident

Shubha D Ojanahalli work on adaptive cruise control and describe about about the design and installation of the Adaptive Cruise Control System (ACC) [1] for two wheelers. The proposed idea is an extension of the same system that has been successfully implemented in four wheelers [1]. Ultrasonic sensor is used to monitor the distance and the signal is being sent to the AT 89c51ED2microcontroller, which controls the speed of the vehicle through brake control and fuel flow control. They develop a device which is comprises of two parts- mechanical part and electronic part. The electronic part mainly consists of sensor, microcontroller and driver circuit. The mechanical part consists of brake system, motors and valve. Here the speed of the controlling the fuel flow to the carburetor and by applying brake to the disc brake of the front wheel.

Figure 2.1 is a prototype made by them in which the serial output coming from the sensor on detection of the vehicle ahead, is given to the microcontroller. The microcontroller is programmed such that the rotation of the stepper motor is different for different ranges of distances detected. For near distance, the stepper motor rotates by 90 degrees in clockwise direction thereby, restricting the amount of fuel flow. For very near distance, stepper motor further rotates to complete 180 degrees providing still more restriction to the flow to reduce the speed. When the vehicle moves away from the host vehicle, stepper motor starts to rotate in anti- clockwise direction to reduce the rotate in restriction to flow thereby, bringing back the speed of the vehicle. The working model of the ACC is shown.

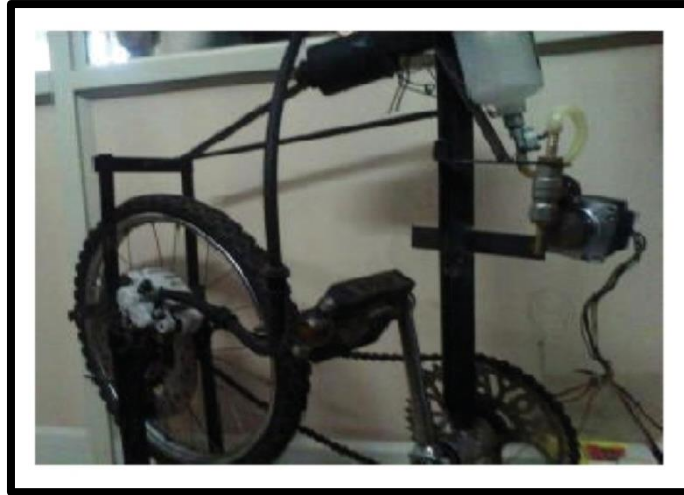


Figure 2.1: Prototype of proposed idea by Shubha D janahalli

Device made by Shubha D janahalli is regarding reducing accident by adjusting the fuel flow in the carburetor, which is totally different as we compare it to our project. So the results of this project is not useful to us, because we are developing the device which maintain vehicle's speed at our desired speed.

2.2) ACC on electric vehicle

D A Purba work on implementation of ACC on electric vehicle. In his study he state that electric motorcycle is one of the vehicles that can operate using alternative fuels, namely with electrical energy. Travel between cities over great distances, of course which results in a high risk of accidents. Additional features are needed that can warn and stop the electric motorcycle before crashing into other vehicles in front of it. Cruise Control is an automatic technology that is embedded in the automotive system to maintain a constant speed. Prototype refers to the physical and digital representation of pre-production models. This research focuses on applying ultrasonic sensors as distance detectors for the prototype adaptive cruise control system which will be installed as an additional feature on electric motorcycles. The method used in this research starts from simulating a series of systems on Proteus 7.9 software, then designing electrical and software devices on prototyping, and testing. The results obtained by adaptive cruise control mode by testing the variation of distance to PWM duty cycle changes in the simulation compared to prototyping, found an error of $\pm 1.53\%$, this shows the prototype Adaptive Cruise Control can work well.

He make a device which is embedded in the automotive system to maintain a constant speed. The cruise control feature is currently developing into Adaptive Cruise Control which is equipped with radar so that it can detect the speed and distance of the surrounding vehicles. Adaptive cruise control, abbreviated to ACC, describes a method of vehicle speed control which adapts to the traffic situation. In this study focused on using an ultrasonic sensor as a distance detection sensor that is US-015 and using *Arduino UNO R3* as the main microprocessor. The method starts from designing a series of Adaptive Cruise Control systems using Proteus 7.9 software, which is an open source electric simulator for circuit design. Figure 2.2 shows the layout of simulation circuit done by using Proteus 7.9 software.

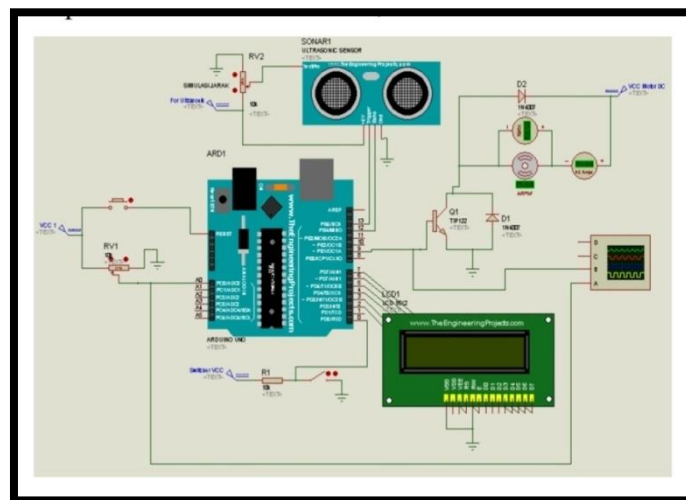


Figure 2.2: Simulation circuit on Proteus 7.9 software

It seen that test carried out by D APurba are valid and the system may be applicable to electric motorcycle as an additional feature by adjusting the real conditions. In this research we get some idea about use and application of Arduino UNO R3. So we also selected Arduino UNO R3 as microprocessor. And also get some idea about simulation of electronic circuit by using Proteus 7.9 software.

2.3) Cruise control in motorcycle

By Pratik PatoleCruise Control is a tech available for many a cars and a few rare bikes. The basic use of cruise control is to keep your vehicle running on a steady speed, irrespective of the conditions and the terrain. It is actually a computerized unit which needs to be paired with your bike's ECU so that the data can be fed accordingly and the speed can be controlled likewise. Now theoretically cruise control can be fit on almost all bikes, but for the best results the bare minimum requirements for cruise control are Fuel Injection and Ride by Wire technology. These two have been covered in our previous articles, so do refer them for more information on the same. Now fitting of a cruise control is a really complicated task, so I am not going into complete details of those here. But just to name the basics, the cruise control gets a control unit which you have to set on the top of the bike so that you can manage the settings. The main backend unit of the cruise control has to be set on the inside of the bike depending on where you get space, but always as away from the engine as possible. Then there are lots of wires that go into it which has to be connected to the ECU, the Speedometer and the FI unit as well.

When the cruise control kicks in, there is a need to calibrate it according to the bike's power output, the torque curve and the rest of it so that the cruise control's computer is in proper sync with the throttle input, the speedometer reading and the ECU to control the bike at a particular speed. When all this is done, you are good to go cruising on long hauls without straining your wrist.



Figure 2.3: Controller of cruise control

As describe in this article, cruise control is only available in high end petrol vehicle. The controller is very complex to use as there is On button, Off button, Engage button and Disengage button. This not makes the device user friendly and somewhat difficult to operate. By seen such a situation we design a controller in such a way that, it only contain one button to engage/activate the cruise control. And it will disengage automatically when the brake is applied.

2.4) Dynamic Radar Cruise Control and Lane-Keeping Assist

We take this study for only knowledge point of view to survey another car systems .Advanced crash avoidance and driver assistance technologies potentially can prevent or mitigate 14 many crashes. Previous surveys with drivers have found favorable opinions for many advanced technologies; 15 however, these surveys are not necessarily representative of all drivers or all systems. As the technologies spread 16 throughout the vehicle fleet, it is important to continue studying driver acceptance and use of them. Method: This 17 study focused on 2010–2013 Toyota Sienna and Prius models that were equipped with adaptive cruise control, 18 forward collision avoidance, and lane departure warning and prevention (Prius models only). Telephone inter- 19 views were conducted in summer 2013 with 183 owners of vehicles with these technologies. Results: About 9 20 in 10 respondents wanted adaptive cruise control and forward collision avoidance on their next vehicle, and 21 71% wanted lane departure warning/prevention again. Males and females reported some differences in their 22 experiences with the systems; for example, males were more likely to have turned on lane departure warning/ 23 prevention than females, and when using this system, males reported more frequent warnings than did females. 24 Relative to older drivers, drivers age 40 and younger were more likely to have seen or heard a forward collision 25 warning.

Conclusions: Consistent with the results in previous surveys of owners of luxury vehicles, the present 26 survey found that driver acceptance of the technologies was high, although less so for lane departure warning/ 27 prevention. Experiences with the Toyota systems differed by driver age and gender to a greater degree than in 28 previous surveys, suggesting that the responses of drivers may begin to differ as crash avoidance technology 29 becomes available on a wider variety of vehicles. Practical application: Crash avoidance technologies potentially 30 can prevent or mitigate

many crashes, but their success depends in part on driver acceptance. These systems will be effective only to the extent that drivers use them.

2.5) Adaptive Cruise Control System for an Electric Vehicle

With the depletion of fossil fuels, and the increased environment concerns, there is more and more interest in Electric Vehicles (EVs) from automobile manufacturer and governments in the entire world. In this paper, a fractional order model reference adaptive control (FOMRAC) system is designed for the cruise control of a DC motor-driven electric vehicle. The vehicles power train model is exposed and the control design is developed and detailed. The major contribution in this work is to use fractional order reference adaptive controllers for both two layer control loops: regulating the current for the inner loop and the desired speed tracking for the outer. Numerical simulations illustrate the performance improvement using the proposed control strategy.

The electric vehicle is a vehicle whose propulsion is provided exclusively by one or more electric motors. It can draw its energy from embedded resources like a battery, or be connected to an external source, for example via a centenary. The engine can be loaded, as in the case of most land vehicles, or outside as in the case of cable transport. Nowadays, electric vehicles become more and more popular; the sale of them has progressed in the world for the reason that electric propulsion has a better efficiency than combustion engines; they need very simple maintenance, requiring very little spare parts.

Many techniques have been used for driving control of EVs, for instance, in (Osman et al., 2009), PID and state space are designed for the cruise control system.

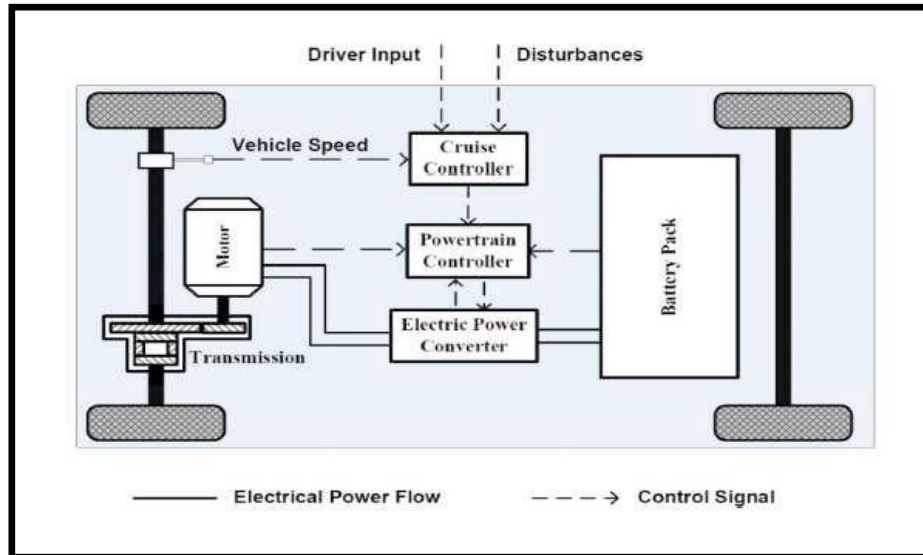


Figure 2.5: EV power train including cruise control system.

SR.NO	AUTHOR	TITLE OF RESEARCH PAPER/BOOK	FINDING/CONCLUSION
1	Pratik Patole	BikersMedia	Cruise control is expensive. But then there are various aftermarket fits available as well. Sadly there are really rare offerings available in India. (Fig: 2.1)
2	Shubha D Ojanahalli (B.VBhoomaraddi college of Engg, Hubli, India')	Adaptive cruise control system for two wheelers to avoid and reduce accidents	Vehicle's speed is decelerated through the brake control. Again the vehicle speed is accelerated back to initial speed once the leading vehicle moves away from the vehicle defined range. (Fig: 2.2)

Design and Manufacturing Adaptive Cruise Control System.

3	D.A. PURBA	Prototyping Adaptive Cruise Control on Electric Motorcycle	The cruise control feature is currently developing into ACC which is equipped with radar so that it can detect the speed & distance of surrounding vehicles. Adaptive cruise control abbreviated to ACC describes a method of vehicle speed control which adapts to the traffic situation. (Fig: 2.3)
4	Angela H. Eichelberger, Anne T. McCart	Toyota drivers' experiences with Dynamic Radar Cruise Control	Overall, Toyota's Dynamic Radar Cruise Control has proven to be a valuable feature for many drivers, reflecting the continuous efforts in the automotive industry to enhance both safety and driving comfort through innovative technologies
5	HananeBalaska,SamirLadaci, Horst Schulte ,AbdelbakiDjouambi	Adaptive Cruise Control System for an Electric Vehicle	Adaptive Cruise Control (ACC) system for electric vehicles represents a significant step forward in enhancing both the convenience and efficiency of driving experiences. Electric vehicles become more prevalent, the integration of ACC advancing smart and sustainable transportation solutions, bringing us closer to a future of efficient, eco-friendly driving.(Fig : 2.5)

(Table no 2.6: LITRATURE SURVEY CONCLUSION / FINDINGS)

2.7) Selection of Raw Material

We are making prototype of the device, so we have manufacture the components. The best way to manufacture it is by using 3D printing technology. But while 3D printing material selection is very important step, because the material is selected in such a way that it have to sustain the given load as per requirement. Here we have to sustain 30 N of load of carburetor spring, so we selected Ploylactid Acid (PLA) material for printing the components.

A Nugroho, R Ardiansyah, L Rusita and IL Larasati (Aeronautics Technology Center – LAPAN, Indonesia) described the effect of different layer thickness of PLA by 3D printing on flexural properties. Additive manufacturing, also known as 3D printing, is a common material extrusion process using (bio) polymers PLA. The study begins from manufacture the solid 3D model based on the ASTM standards for flexural properties test of the material using Three-point bending method. Three-point bending test was conducted with Tension Universal Testing Machine - AND RTF-2410 with a 100kN load cell, specimen shape and size according to standard size ASTM D 790. The result shown that the layer thickness had an effect on flexural strengths of PLA samples. The maximum flexural strengths from $L_t = 0.4$ to 0.5mm were significantly increase. Moreover, it is worth nothing that ductility decreased as layer thickness increased. According to test result that the maximum flexural strength occurred at 0.5 mm layer thickness with 59.6 MPa and the minimum flexural strength occurred at 0.1 mm layer thickness with 43.6 MPa. The higher layer thickness tended to promote higher strength. The thicker layer is the stronger layer bond in holding the load bending. In this study the thicker layer have tended to shown a 90-degree delamination fracture and the thinner layer have tended to shown a 45-degree delamination fracture according to the direction of printing is $\pm 45^\circ$.

Mechanical Properties of PLA	
Properties	PLA
Tensile strength (Mpa)	15.5 – 72.2
Tensile Modulus (Gpa)	2.020 – 3.550
Elongation at break (%)	0.5 – 9.2
Flexural modulus (Gpa)	2.392 – 4.93

(Table No 2.7: Mechanical properties of PLA)

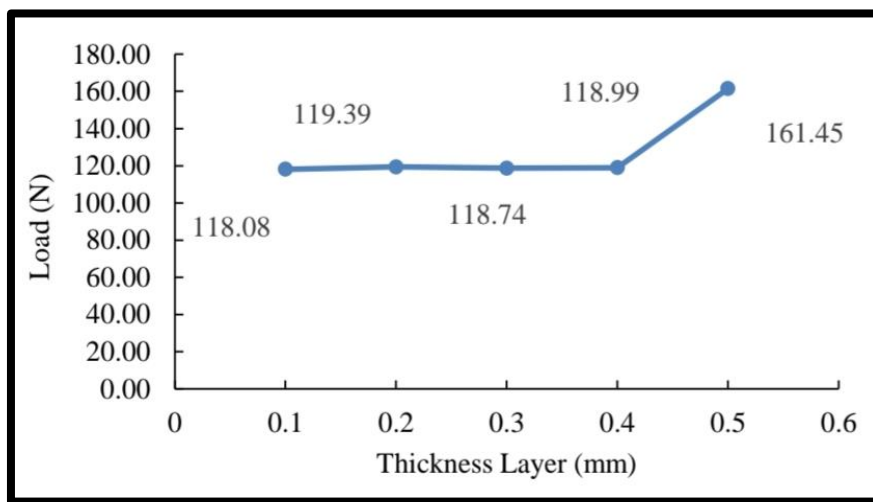


Figure 2.7.1: Graphical comparison of average maximum Load for the Flexural specimen as a function of layer thickness.

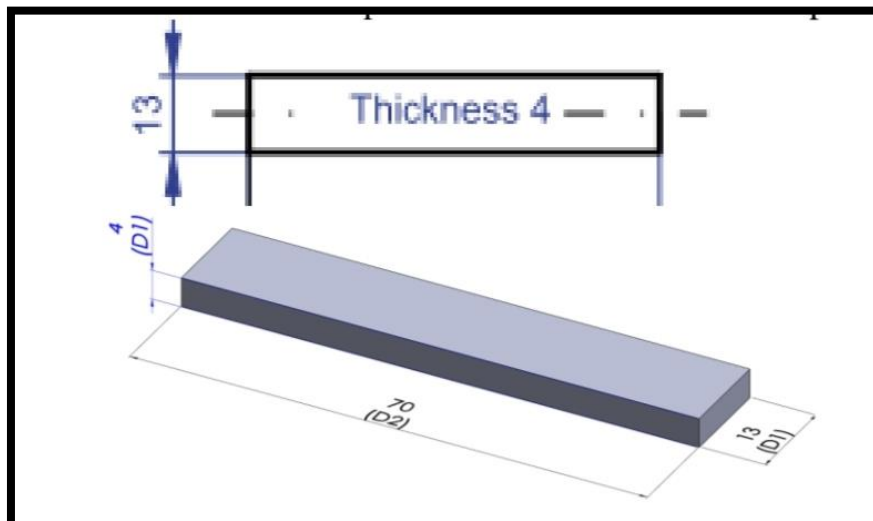


Figure 2.7.2: Layer thickness

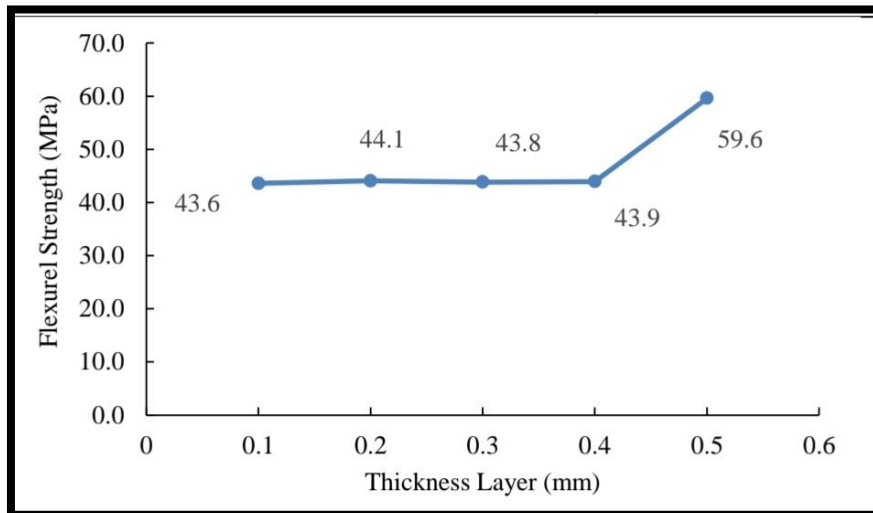


Figure 2.7. 3: Graphical comparison of average maximum flexural strength (of) for the flexural specimens as a function of layer thickness

Figure 2.7.3 shows that the increase in layer thickness affects the flexural strength of PLA materials made with 3D printers. At layer thickness {0.1, 0.2, 0.3 and 0.4} mm there is no difference but in the 0.5mm layer thickness there is a significant increase from 43.9MPa to 59.6MPa.

CHAPTER 3

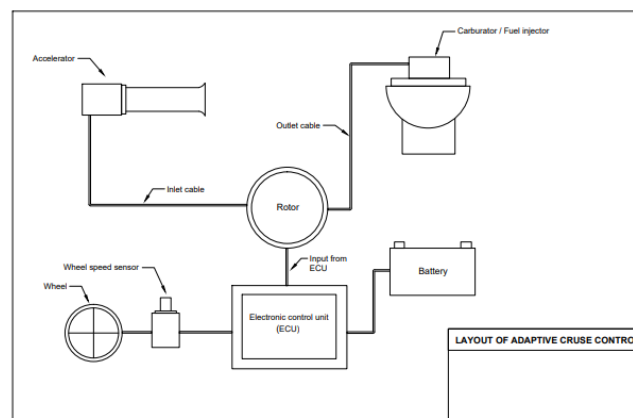
PROPOSED EXPERIMENTAL SETUP

3.1) Planning & Design Layout:

Stepwise explanation of Planning of layout

1. There will be inlet cable connected to Accelerator and rotor mechanism.
2. Secondly the outlet cable is connected through rotor mechanism to carburetor.
3. There next to the wheel wheel speed measuring sensor is provided or place for measuring the rpm of the wheel shaft.
4. We place ECU (Electronic Control Unit) for taking data from wheel speed sensor to operate rotor mechanism for working of adaptive cruise control system of the bike.
5. We placed battery for the giving energy to the ECU unit.
6. Develop a control algorithm that governs the ACC system's behavior. This algorithm should take into account the data from sensors, the desired speed set by the driver, and safety margins to determine the optimal speed for the vehicle.
7. So for the working of our project there will be systematic layout for understanding the design and working plan as follows:

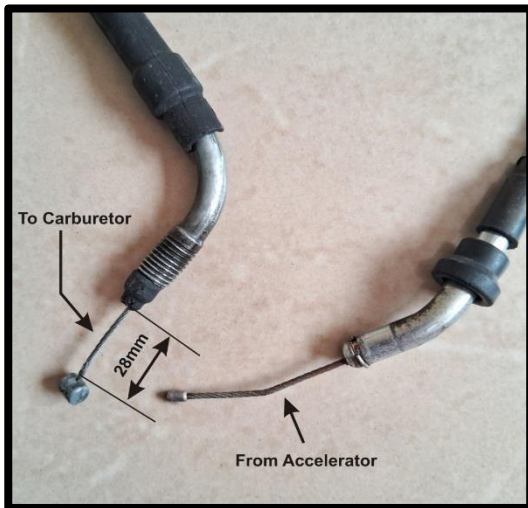
3.2) Design Layout:



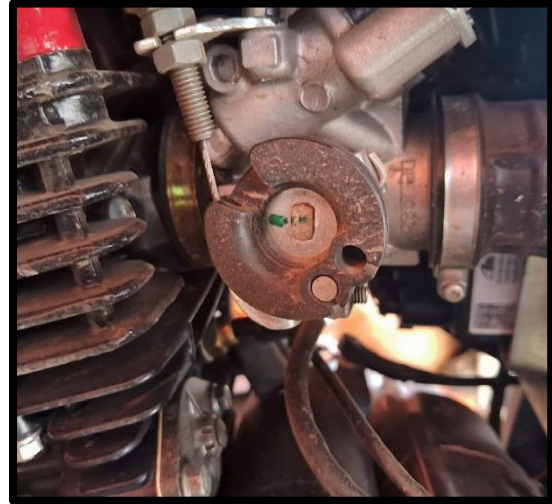
(Figure no: 3.2 Layout of system)

3.3) Real word reference take to develop design of rotor & inlet/outlet cable in

Layout:



(Pic no: 3.3.1 I/O Cable)



(Pic no : 3.3.2 Rotor mechanism)

3.4) Material Selection of spring & slider Actual taken calculations:

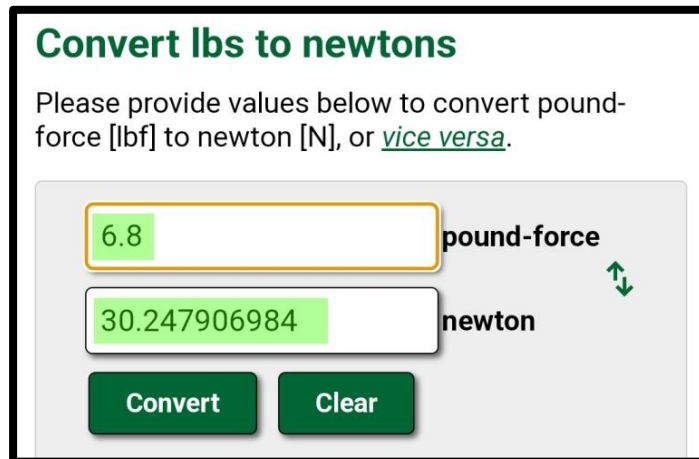


(Pic: 3.4.1. Spring and slider)

Throttle Return Spring - 1.75" x 0.250" x 0.034"		
Throttle Return Spring Specifications		
Free Length	Outside Diameter	Maximum Load
1.75"	0.25"	6.8 lb

<https://www.pegasusautoracing.com> > ...

(Pic: 3.4.2. Throttle return spring specifications)

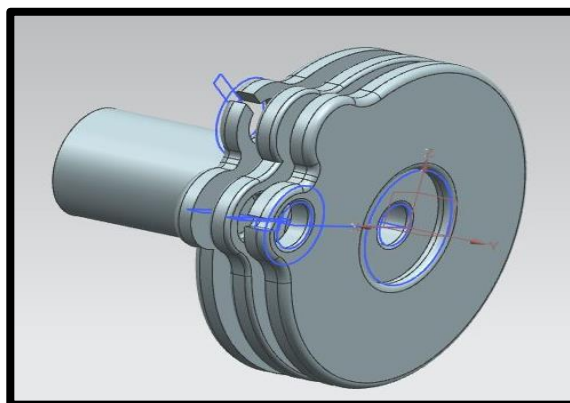


(Pic: 3.4.3 Conversion of lb. to Newton)

3.5) Design of Rotor Mechanism throttle shaft:

1. Rotor mechanism is main component of our system.
2. In rotor mechanism there is throttle shaft which is design in such a way that, inlet cable from the accelerator is attached to the 1st disc of the shaft where as another cable which goes to carburetor is attached to the 2nd disc which is behind 1st disc. There is one rotary motion present in shaft.
3. This shaft is made up of PLA material which can bear about 60 Mpa load.
4. Ratchet gear is placed at the end of the shaft where locking of the shaft takes place.
5. This shaft is cover under casing made up of same material PLA.

3.6) Design of Rotor Mechanism throttle shaft using the UGNX 10 software 3D Model.

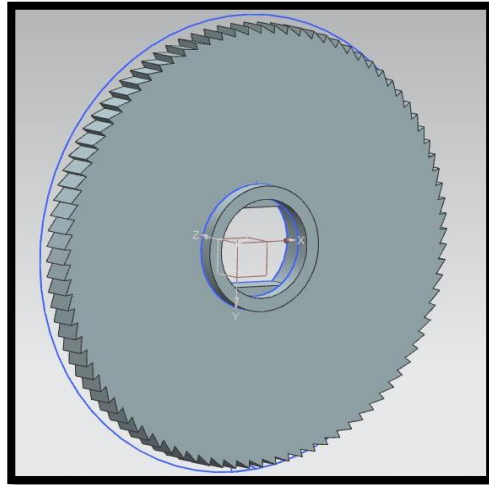


(Fig no: 3.6 Rotor Mechanism throttle shaft)

3.7) Planning & Design of Ratchet mechanism:

1. We use ratchet gear to provide locking of the shaft. This gear is placed at the end of the gear.
2. It consists of a PAWL which locks the gear as it rotates in anti-clock wise direction and were allows it to rotate when clockwise direction.
3. This gear consists of 90 teeth, angle between each teeth is about 4 degrees.
4. This gear is design in such a way that, even in minimum twisting of shaft it provides multiple locking of gear. In 90 degrees of rotation there are 22 teeth provided for locking action.
5. It is made up of same material 'PolyLactid Acid (PLA)'.
6. Width of the gear is 4mm which is sufficient to bearing 30 N of axial load.

3.6) Design of Ratchet mechanism using the UGNX 10 software 3D Model.



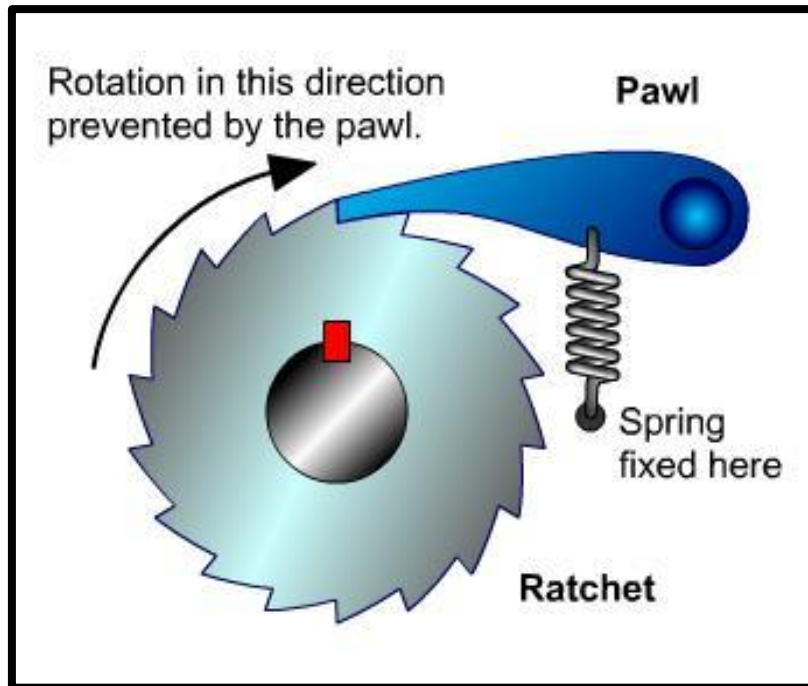
(Fig no:3.6.1 Ratchet mechanism 3d model)

3.7) Locking mechanism use in Rotor:

1. The locking mechanism is an integral component of the rotor system, providing the necessary stability, safety, and precision required for optimal performance.

Ongoing advancements in design and technology aim to further improve the efficiency and reliability of this critical element in rotor assemblies.

2. The primary function of the locking mechanism is to prevent unintended movements of the rotor during critical phases of operation.

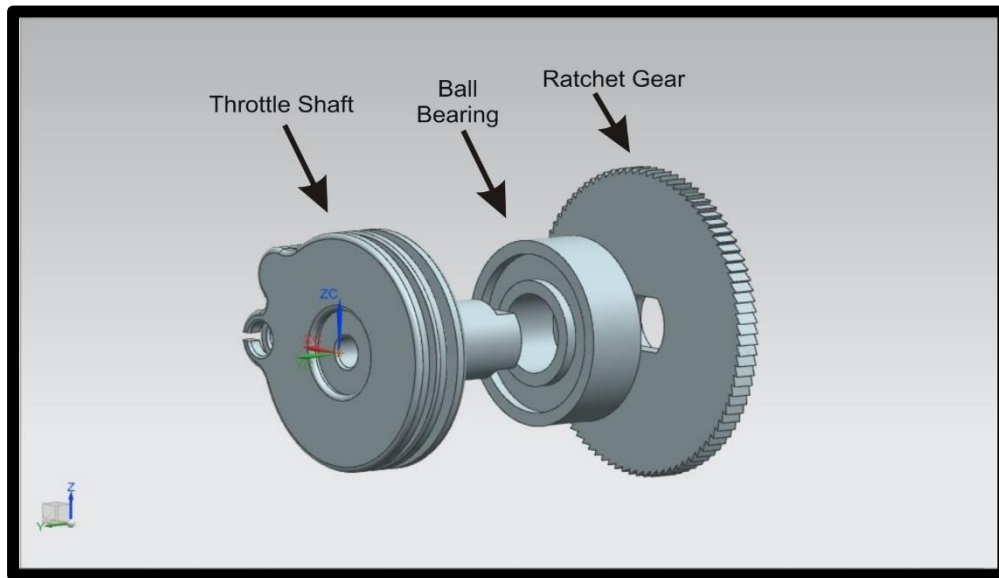


(Fig no:3.7 Ratchet and pawl mechanism)

3.8) Assembly of Throttle shaft, Ball bearing and Ratchet gear:

1. The assembly process of the throttle shaft, ball bearing, and ratchet gear has been successfully completed, demonstrating a cohesive integration of these essential components.
2. The combination of the throttle shaft, ball bearing, and ratchet gear is designed to facilitate smooth and efficient operation within the engine system, contributing to overall performance.

3. The assembly has been designed with maintenance considerations in mind, allowing for ease of routine maintenance and addressing any potential issues that may arise during the operational life of the throttle shaft unit.
4. This assembly is created on the 3d design software known as UGNX 10 by using 3d modeling are as following:



(Fig no: 3.8 3D Model of Assembly created using 3d modeling software UGNX 10)

3.9) Planning for Electronic control unit (ECU):

1. We use Arduino UNO R3 as a ECU for our project to handle the electronic part for the project system.
2. The Arduino Uno proved to be a viable option for implementing a basic Electronic Control Unit, showcasing its versatility, cost-effectiveness, and educational value. As technology advances, exploring more powerful microcontrollers or dedicated ecumay address scalability concerns while maintaining the principles learned through this project.
3. The Arduino Uno provided a platform for customization and flexibility in programming, allowing for the adaptation of control strategies based on specific project requirements

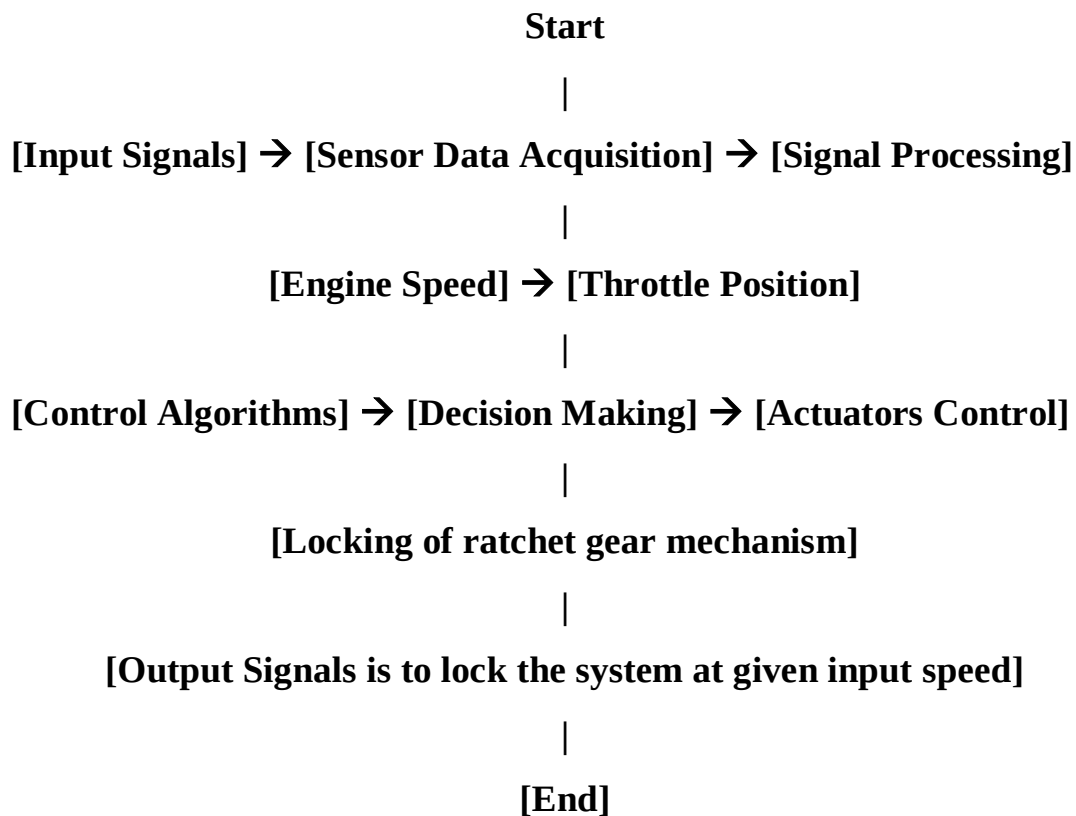
Design and Manufacturing Adaptive Cruise Control System.

4. For the programming we use the user-friendly Arduino IDE platform to made programming the ECU straightforward with limited or basic programming experience.
5. In This project we successfully demonstrated the feasibility of using the Arduino Uno as an Electronic Control Unit (ECU).



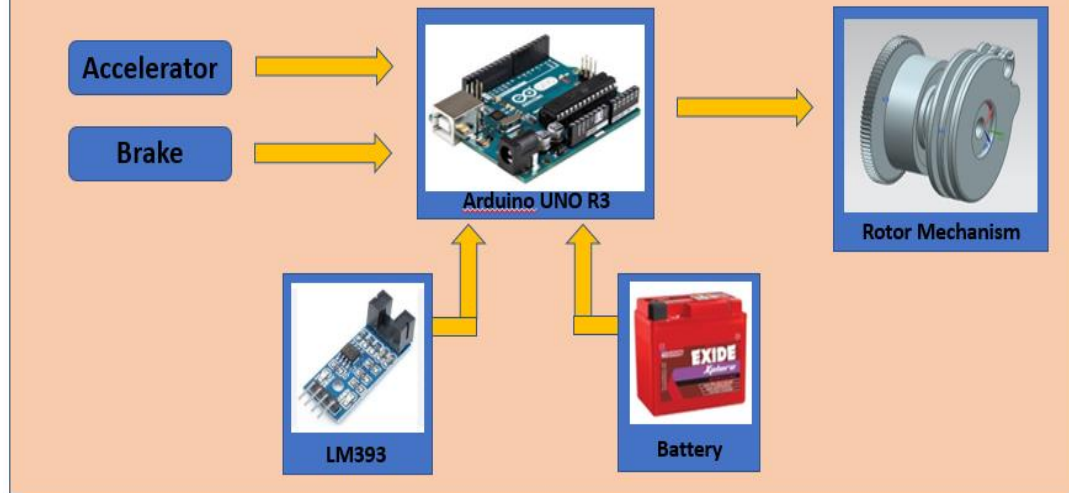
(Fig no:3.9 Electronic control unit (ECU) Arduino UNO R3)

3.10) Working of Electronic control unit (ECU) in our project By Flow Chart:



(Fig3.10:Flow chart by visual of ECU working system in project)

• **Electronic control unit (ECU)**



3.11) Arduino Programming:

```
#include <Servo.h>

const int ledPin = 13;          // LED connected to digital pin 13
const int speedSensorPin = 2;   // LM393 speed sensor connected to digital pin 2
const int buttonPin = 3;        // External button connected to digital pin 3
const int servoPin = 9;         // Servo motor connected to digital pin 9
const int rpmDisplayPin = 10;   // RPM display connected to digital pin 10

volatile unsigned int rpm;
volatile unsigned long lastTime;
volatile unsigned long currentTime;
volatile boolean flag;

Servo myServo;

void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(speedSensorPin, INPUT);
  pinMode(buttonPin, INPUT_PULLUP);
  pinMode(rpmDisplayPin, OUTPUT);

  attachInterrupt(digitalPinToInterrupt(speedSensorPin), calculateRPM, FALLING);
  attachInterrupt(digitalPinToInterrupt(buttonPin), rotateServo, FALLING);

  myServo.attach(servoPin);
  myServo.write(90); // Initialize the servo at 90 degrees
  delay(1000);       // Wait for servo to reach the initial position
}
```

(Figure 3.11.1: Arduino coding)

Line 1: const int ledpin = 13;

LED connected to digital pin 13

Line 2: const int speedsensorpin = 2;

Lm393 speed sensor connected to digital pin 2

Line 3: const int button pine = 3;

External button connected to pin 3

Line 4: const int servo pin = 9;

Servo motor connected to pin 9

Line 5: const int rpmdisplaypin = 10;

RPM display connected to digital pin 10

Line 6: volatile unsigned int rpm;

Typecast rpm data

Line 7: Volatile unsigned long lasttime;

Describe last time

Line 8: Volatile unsigned long currenttime;

Describe current time

Line 9: Void setup()

Programing start for LED display

Line 10: pinMode(ledpin, OUTPUT);

Ledpin is output

Line 11: pinMode(speedSensorPin, INPUT);

Speed sensor pin input

Line 12: pinMode(buttonPin, INPUT_PULLUP);

Button pin is Input

Line 13: pinMode(rpmDisplayPin, OUTPUT);

Display pin is output

Line 14: attachInterrupt(digitalPinToInterrupt(speedSensorPin), calculateRPM, FALLING);

Speed sensor calculate the rpm and display it

Line 15: attachInterrupt(digitalPinToInterrupt(buttonPin), rotateServo, FALLING);

When button is press servo motor rotates

Line 14: myServo.attach(servoPin);

Servo motor attach to servo pin

Line 15: myServo.write(90);

Initialize the servomotor at 90 degree

Line 16: delay(1000);

Takes delay of 1000 millisec

Line 17: void loop()

Program start for led


```
void loop() {  
    digitalWrite(ledPin, HIGH); // Blink the LED  
    delay(500);  
    digitalWrite(ledPin, LOW);  
    delay(500);  
  
    if (flag) {  
        displayRPM();  
        controlServo();  
        flag = false;  
    }  
}  
  
void calculateRPM() {  
    currentTime = micros();  
    rpm = 60000000 / (currentTime - lastTime);  
    lastTime = currentTime;  
    flag = true;  
}  
  
void displayRPM() {  
    digitalWrite(rpmDisplayPin, HIGH);  
  
    // Code to display RPM on a 2x16 LCD can be added here  
    // You may need to include a library and initialize the LCD in the setup() function  
    // Example: LiquidCrystal lcd(0);  
  
    // For simplicity, let's just print RPM to Serial Monitor  
    Serial.begin(9600);  
    Serial.print("RPM: ");  
    Serial.println(rpm);  
    Serial.end();  
}
```

(Figure 3.11.2: Arduino programming)

Line 18: digitalWrite(ledPin, HIGH);

Blink the LED

Line 19: delay(500);

Delay of 500 Millisec

Line 20: digitalWrite(ledPin, LOW)

OFF the LED

Line 21: delay(500);

Delay of 500 milisec

Line 22: if (flag)

Perfrom a function

Line 23: displayRPM();

Display the rpm

Line 24: controlServo();

Control the servo motor

Line 25: flag = false;

Then the flag is false

Line 26: Void calculaterpm()

Now calculate the rpm

Line 27: currentTime = micros();

Current time RPM in micros

Line 28: rpm = 60000000 / (currentTime - lastTime);

Line 29: lastTime = currentTime;

Line 30: flag = true;

Flag is true

Line 31: void displayRPM()

Program starts for display

Line 32: digitalWrite(rpmDisplayPin, HIGH);

On the display

Line 33: Serial.begin(9600);

Monitor ata data rate of 9600 bits per second

Line 34: Serial.print("RPM: ");

Print the rpm in display

Line 35: Serial.println(rpm);

Line 36: Serial.end();

Disable serial channel

```
    delay(1000);
    digitalWrite(rpmDisplayPin, LOW);
}

void controlServo() {
    if (rpm > 90) {
        myServo.write(60); // Rotate servo motor negative 30 degrees
        delay(1000);       // Wait for servo to reach the position
    } else if (rpm > 20) {
        myServo.write(120); // Rotate servo motor positive 30 degrees
        delay(1000);       // Wait for servo to reach the position
    } else {
        myServo.write(90); // Keep servo at the initial position
        delay(1000);       // Wait for servo to reach the position
    }
}

void rotateServo() {
    myServo.write(60); // Rotate servo motor negative 30 degrees
    delay(1000);       // Wait for servo to reach the position
}
```

(Figure 3.11.3: Arduino Programming)

Line 37: delay(1000);

Delay of 1000 millisec

Line 38: digitalWrite(rpmDisplayPin, LOW);

Digital display disable

Line 39: void controlServo()

Program start for servo motor

Line 40: if (rpm > 90)

If rpm of vehicle is less than 90

Line 41: myServo.write(60);

Rotate servo motor 30 degree anticlockwise

Line 42: delay(1000);

It takes delay of 1000 millisec

Line 43: else if (rpm > 20)

If rpm of vehicle greater than 20

Line 44: myServo.write(120);

Rotate servo motor 30 degree

Line 45: delay(1000);

Delay of 1000 millisec

Line 46: else

Line 47: myServo.write(120);

Keep the servo at the initial position

Line 48: delay(1000);

Time delay of 1000 millisec to reach the position.

CHAPTER 4

EXPECTED CONCLUSION AND FUTURE SCOPE

4) Expected Conclusion:

The introduction of Adaptive Cruise Control in two-wheelers represents a promising advancement in rider assistance technology, having the way for safer, more comfortable, and technologically sophisticated urban commuting experiences. As this technology continues to evolve, it holds the potential to reshape the future of two-wheeler transportation.

In conclusion, the implementation of Adaptive Cruise Control (ACC) in two-wheelers represents a significant step forward in enhancing rider safety and comfort. By incorporating ACC technology into motorcycles and scooters, we aim to provide riders with an intelligent system that can lock the vehicle's speed based on the road conditions and riders command. This not only reduces the mental load on the rider but also contributes to a more relaxed and enjoyable riding experience.

4.1) Future Scope:

1. Future ACC systems may offer customizable riding modes, allowing riders to select preferences based on traffic conditions, weather, or personal riding style.
2. ACC systems can be designed to optimize fuel efficiency and reduce emissions by maintaining optimal speeds and minimizing unnecessary acceleration and deceleration.
3. Future two-wheelers with ACC can feature user-friendly interfaces, such as touchscreens or voice commands, making it easy for riders to customize and control the ACC system.
4. By automating speed adjustments, ACC can reduce the cognitive load on riders. This can lead to less fatigue during long rides, making two-wheeler travel more comfortable

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